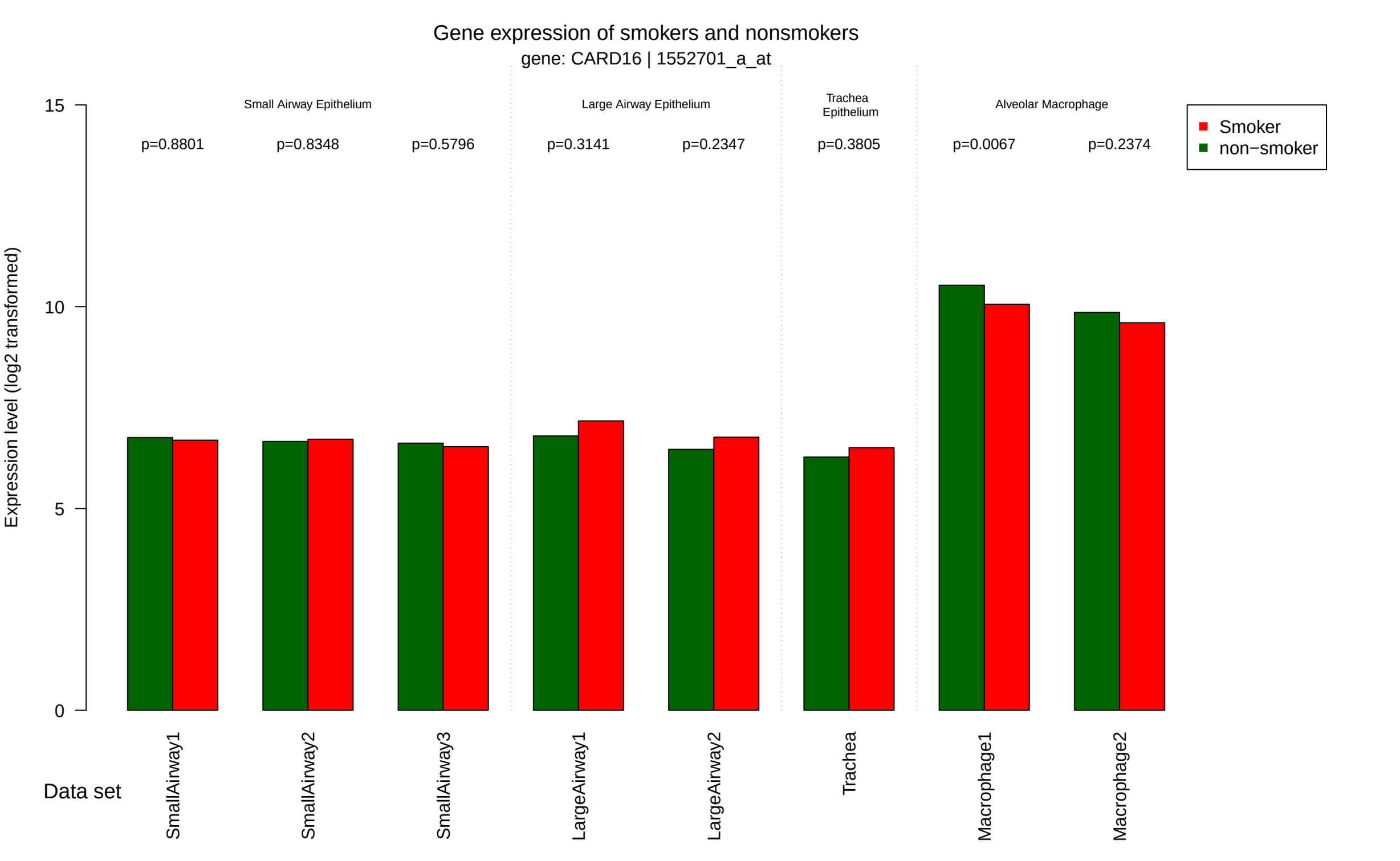
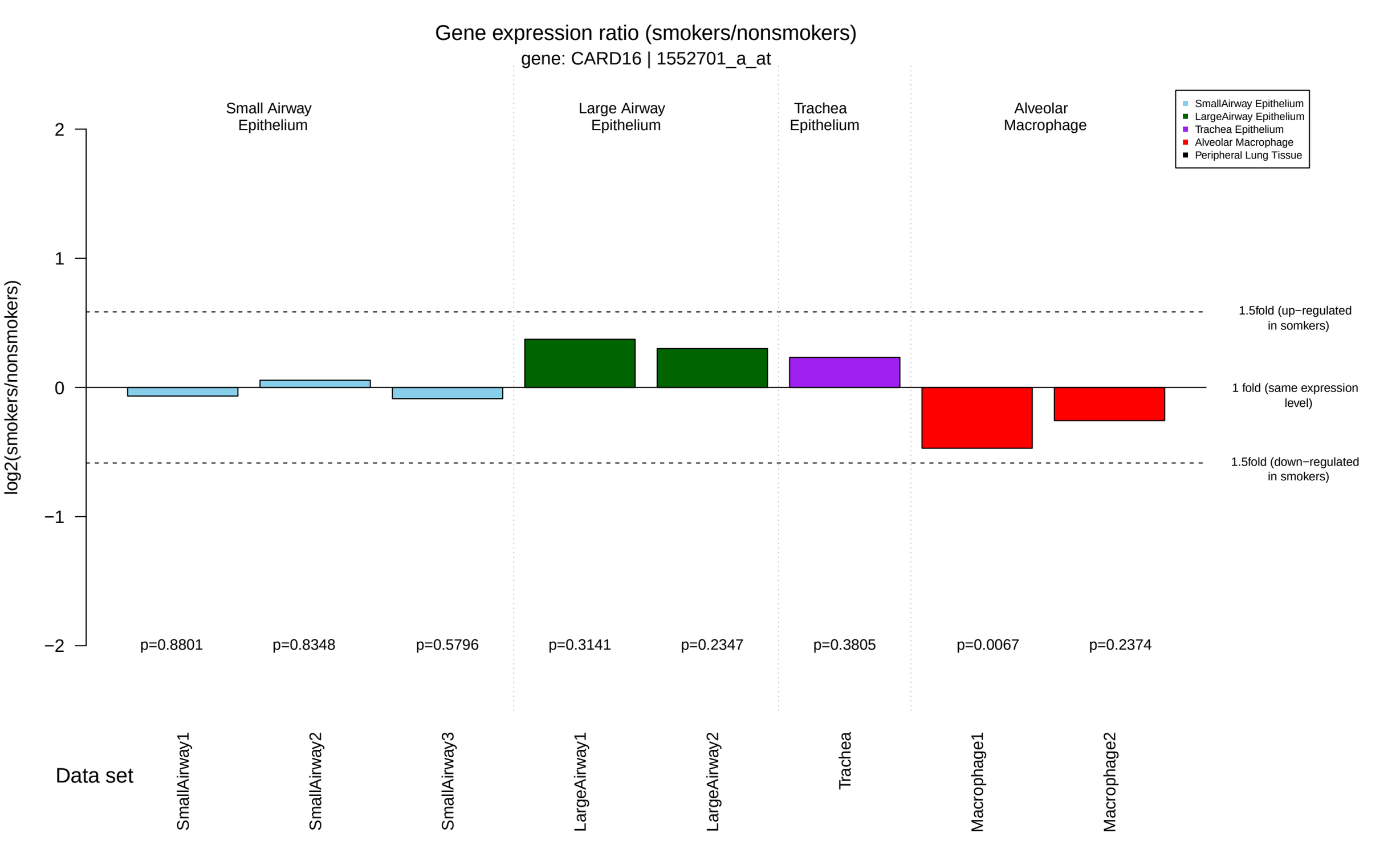
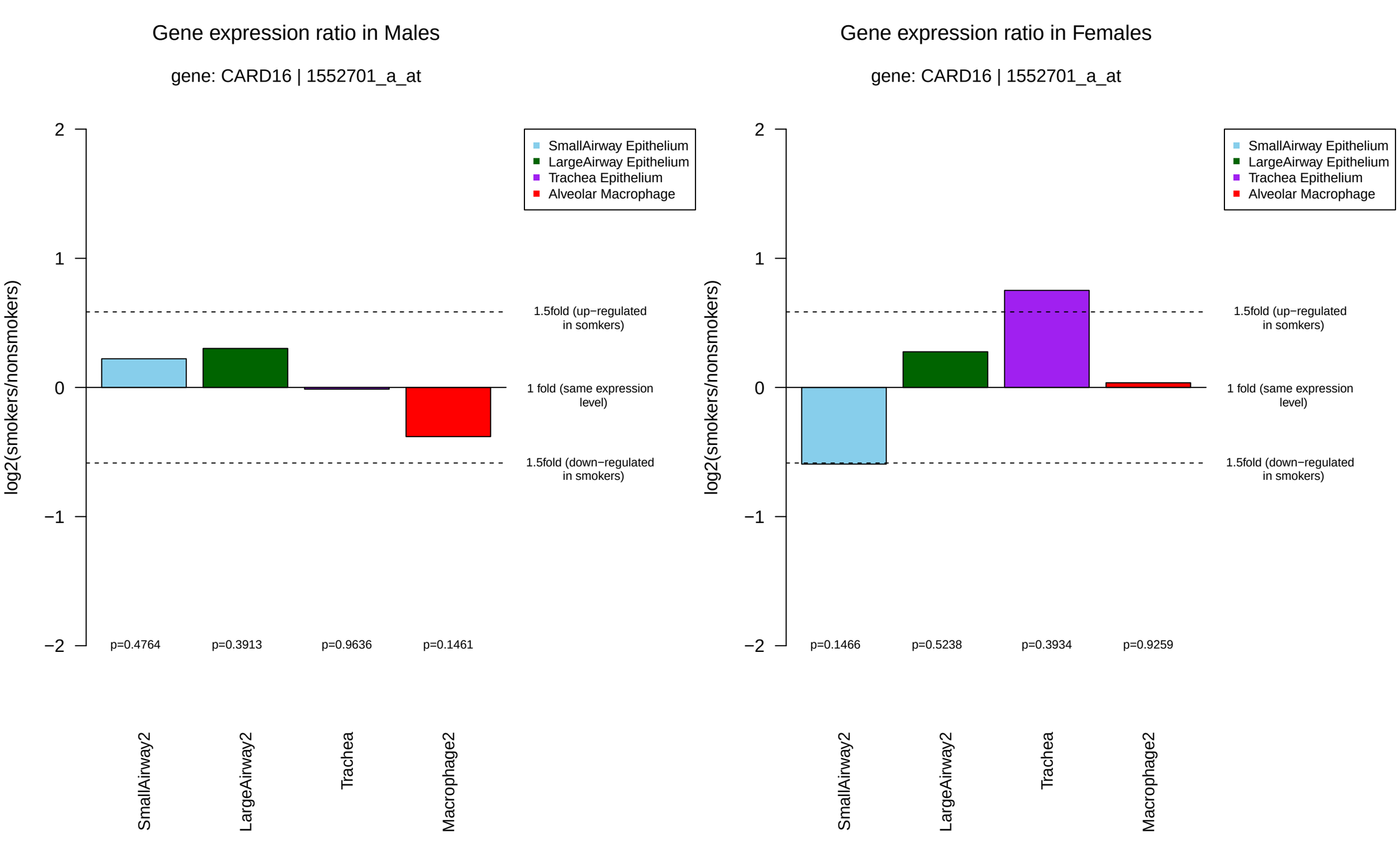
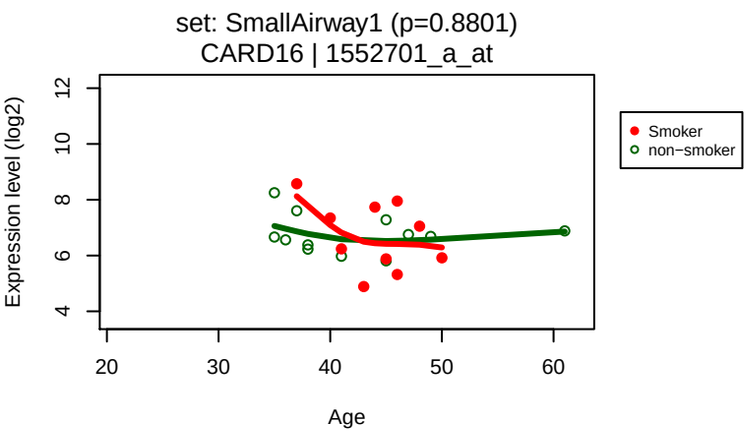
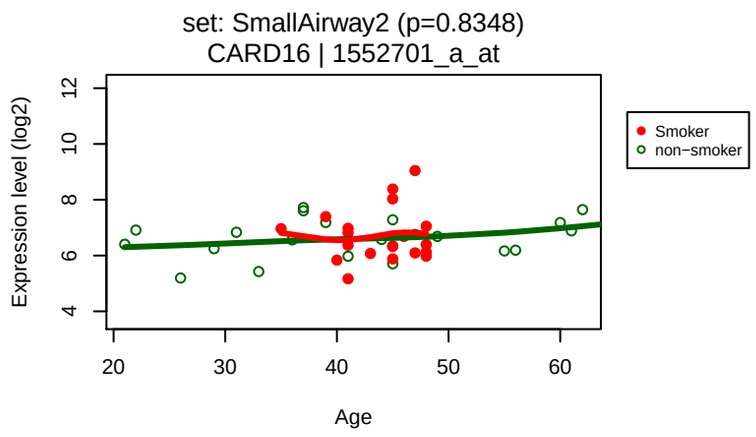
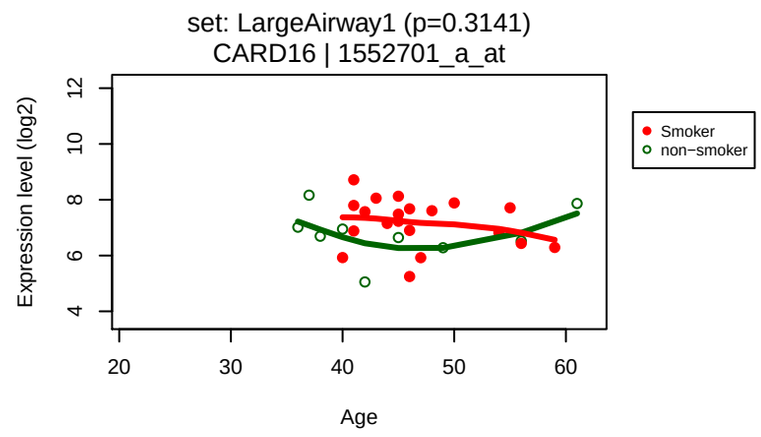

Gene expression of smokers and nonsmokers
gene: CARD16 | 1552701_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8801
p=0.8348
p=0.5796
p=0.3141
p=0.2347
p=0.3805
p=0.0067
p=0.2374
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

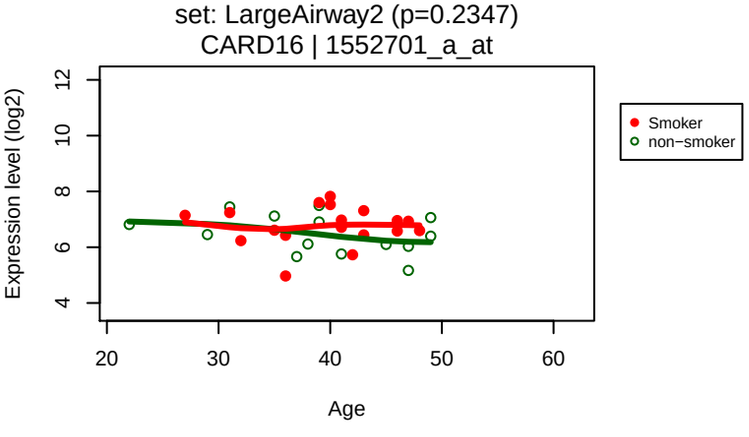

Gene expression ratio (smokers/nonsmokers)
gene: CARD16 | 1552701_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8801
p=0.8348
p=0.5796
p=0.3141
p=0.2347
p=0.3805
p=0.0067
p=0.2374
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

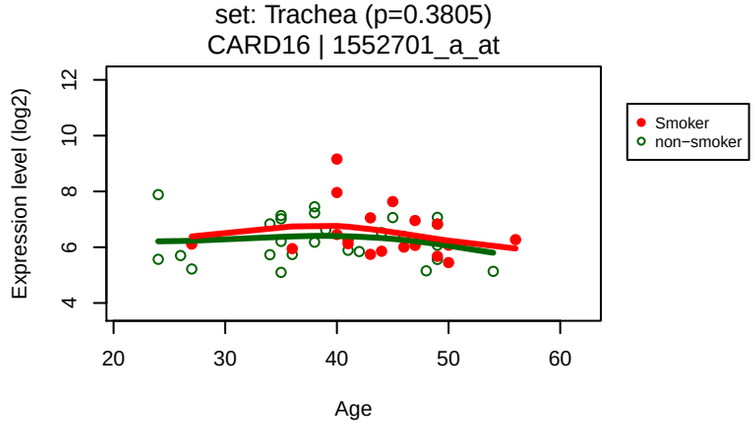

Gene expression ratio in Males
gene: CARD16 | 1552701_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4764
p=0.3913
p=0.9636
p=0.1461
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARD16 | 1552701_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1466
p=0.5238
p=0.3934
p=0.9259
SmallAirway2
LargeAirway2
Trachea
Macrophage2

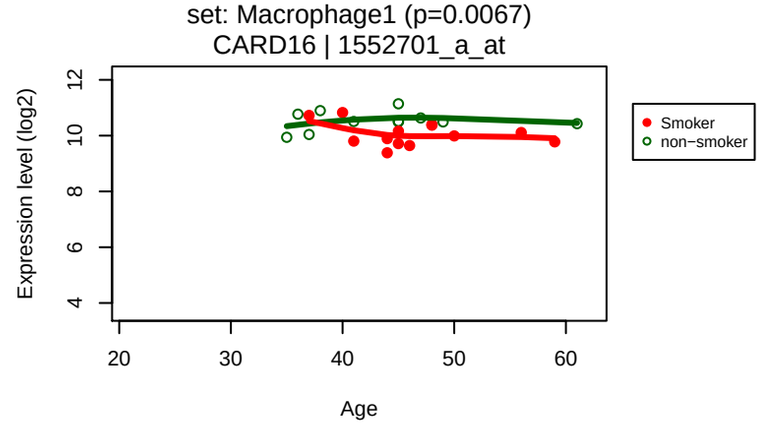

set: SmallAirway1 (p=0.8801)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker

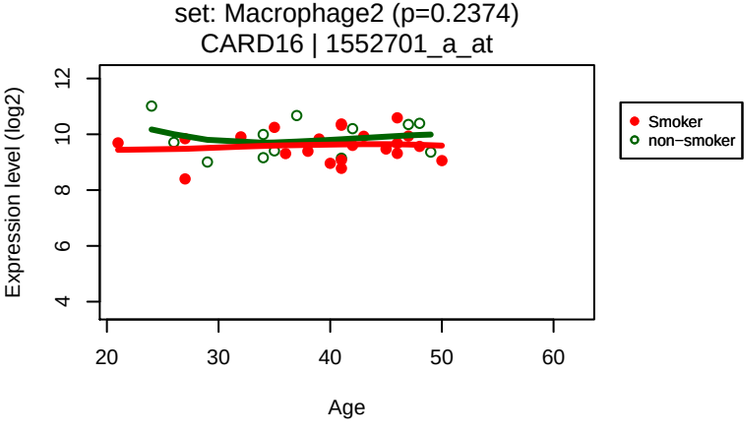

set: SmallAirway2 (p=0.8348)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker

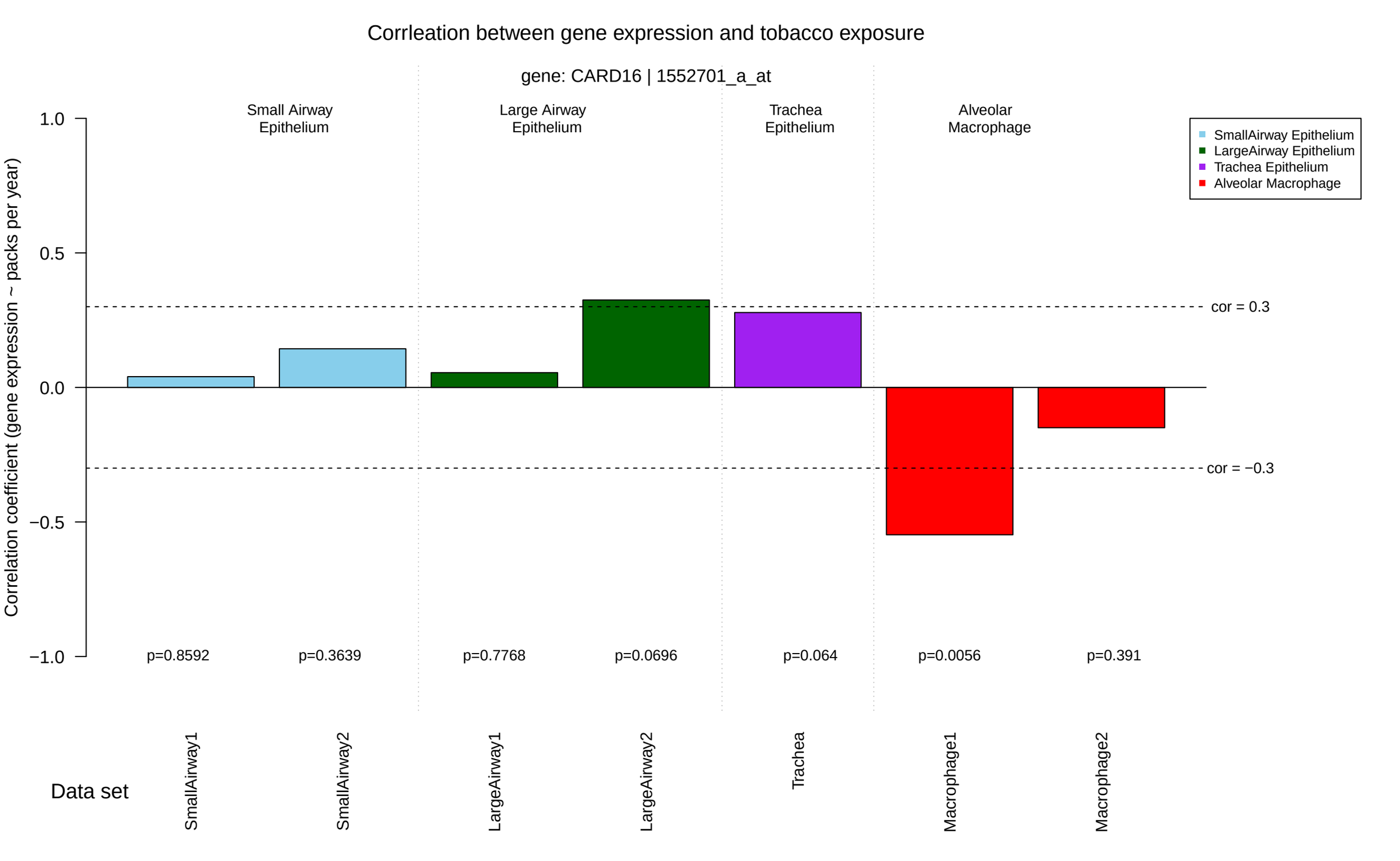

set: LargeAirway1 (p=0.3141)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2347)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3805)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0067)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2374)
CARD16 | 1552701_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARD16 | 1552701_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8592
p=0.3639
p=0.7768
p=0.0696
p=0.064
p=0.0056
p=0.391
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2